Author MARCON MICHAEL - michael.marcon@stellantis.com
Co-Author LE MOINE BRUNO - bruno.lemoine@stellantis.com
Author Department FCA US -- ENG SYS FUEL SYSTEMS - 1670
Co-Author Department FCA ITALY -- ENGINE SYSTEMS - FUEL SYSTEM

GLOBALLY HARMONIZED DOCUMENT

Change level	Date	Description of change
-	25-NOV-2015	Initial release
A	07-JUL-2018	Updated to new template and replaced old specs.
B	04-FEB-2022	Updates made to: 5.1, 5.3, 5.4, 5.5, 5.6, 6, 6.2.4, 6.3.1, 6.3.4, 6.3.5, 6.3.6, 7.5, 8.2, 9.3, 9.5, 9.6, 9.7, and Annex A.

For information, please contact author and co-author in the lateral label of this document

For APAC, LATAM and Maserati contacts, please consult SD.001300

on December 01, 2023 09:03 AM (UTC+1) - userid: T7284G0
PRS_STLA_CHINA@stellantis.com

ANY PRINTED COPY IS TO BE DEEMED AS UNCHECKED; THEREFORE THE UPDATED COPY MUST BE CHECKED IN THE APPROPRIATE WEB SITE

CONFIDENTIAL
THIS DOCUMENT MUST NOT BE REPRODUCED OR CIRCULATED TO THIRD PARTIES WITHOUT PRIOR WRITTEN CONSENT BY THE RELEVANT FCA COMPANY.
IN CASE OF DISPUTE, THE ONLY VALID REFERENCE IS THE ENGLISH EDITION

QUICK CONNECT – FUEL SYSTEMS

TABLE OF CONTENTS

1 GENERAL/INTRODUCTION 3
1.1 Purpose 3
1.2 Component/Commodity Description 3
1.3 Application 4
1.4 Hazardous Material Control 4
1.5 Mandatory, Recommended or Specified Tests 4
1.6 Test Planning 4
1.7 Suggested, Recommended or Specified Components and/or Suppliers 4
1.8 Auxiliary device (Secondary Latch) required characteristics: 5
2 REFERENCES 5
3 DEFINITIONS/ABBREVIATIONS/ACRONYMS/SYMBOLS 6
4 SPECIAL TEST EQUIPMENT 7
4.1 Test environment (unless otherwise specified) 7
5 ENVIRONMENTAL REQUIREMENTS 7
5.1 Corrosion 7
5.2 Zinc Chloride Resistance 7
5.3 Calcium chloride resistance (EMEA Only) 8
5.4 Chemical Exposure 8
5.5 Internal Fuel Resistance Test 8
5.6 Storage 10
6 APPEARANCE / PHYSICAL / MECHANICAL / ELECTRICAL REQUIREMENTS 10
6.1 Appearance Requirements 10
6.2 Physical Requirements 11
6.3 Mechanical Requirements 12
6.4 Electrical Requirements 18
7 FUNCTIONAL REQUIREMENTS 18
7.1 Leak Test<S,E> 18
7.2 Assembly Effort<D> 19
7.3 Pull-Apart Effort<D> 20
7.4 Side Load Capability 20
7.5 Flow Restriction 21
7.6 Gravel resistance test 22
8 SAFETY AND REGULATORY REQUIREMENTS 22
8.1 Vehicle Impact 22
8.2 Hydrocarbon Permeation Emissions<E> 23
9 RELIABILITY / DURABILITY REQUIREMENTS 23
9.1 System and Vehicle Level Requirement 23
9.2 Design Failure Modes and Effects Analysis (DFMEA) 23
9.3 Latch Fatigue and On/Off Insertion for Serviceability 23
9.4 Life Cycling 24
9.5 Rocker 24
10 PRODUCT ASSURANCE 25
10.1 Approved Best Practices 25
11 WARRANTY RETURNED PARTS TEST AND ANALYSIS PROCEDURES 25
Annex A 26

On December 01, 2023 02:23 AM (UTC+1) - userid: T7284G0
PRS_STLA_CHINA@stellantis.com

1 GENERAL/INTRODUCTION

1.1 Purpose

This standard covers the environmental, physical, electrical, mechanical, functional, regulatory and durability requirements for Quick Connect – Fuel Systems.

1.2 Component/Commodity Description

The fuel system quick connect fitting is comprised of a case, o-rings, spacers, and a retaining clip. The fitting functions as a joining device and seal for tubing end-forms to FCA specified hose/tube material in **Figure 1** - **Definition of Quick Connector**.

Quick connect couplings function by joining the connector to a mating tube end form then pulling back to assure a complete connection. The requirements stated in this document apply to new connectors in assembly operations unless otherwise indicated. This standard also applies to Integrated Female (Cartridge) Connectors. **Figure 2 - Definition of Cartridge Connector**.

The female connector shall interface to a male connector per **SAE J2044**. The line connection can be either a barbed connection or a male quick connect fitting. If it is a male quick connect fitting it shall meet the requirements of **SAE J2044**.

If the quick connector is shipped with a shipping plug, the plug must not have interference to the retaining latch in order to prevent poor retention to the male endform. The plug must not cause deformation to the internal o-rings of the quick connector and be at the minimum tolerance of the male connector, refer to **SAE J2044** dimension "A".

If the connection is a barbed connection, the supplier of the quick connector is responsible to ensure the line connection mates to the quick connector and meets the requirements of this document. Fuel lines (typically nylon multilayer, but may be other constructions) are defined by **SAE J2260**.

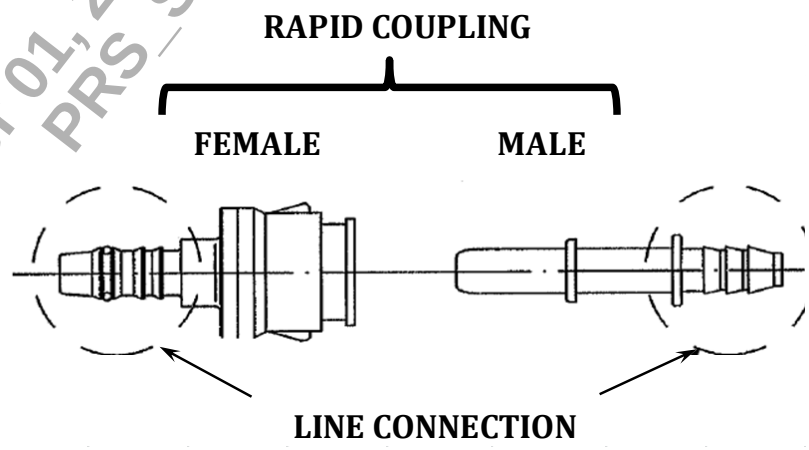

Figure 1 - Definition of Quick Connector

on December 01, 2023 02:23 AM (UTC+1) - userid: TZ284G0
PRS_STELLA_CHINA@stellantis.com

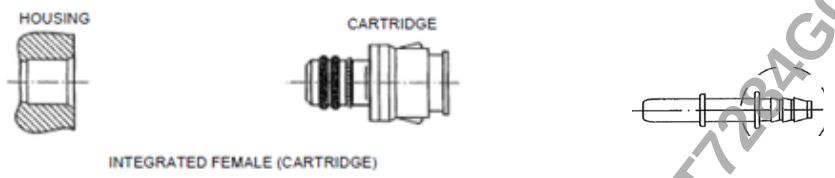

Figure 2 - Definition of Cartridge Connector

1.3 Application

This standard is limited in its application to those CAD files, drawings or engineering illustrations which call out this standard number, or refer to this standard within some other standard or specification. In order to comply with this specification, the joints must have demonstrated the same performance levels for each type of utilized pipe (rubber, polyamide, coextruded etc.).

1.4 Hazardous Material Control

Products and processes used to manufacture the products listed in this standard must conform to the employee and consumer health, employee safety, regulated substances, recycling and environmental reporting requirements.

1.5 Mandatory, Recommended or Specified Tests

Mandatory or suggested tests if any provided within this standard are meant to assure compliance to the specification but do not relieve the responsible entity from ensuring that the component or system meets all environmental, physical, electrical, mechanical, functional, regulatory and durability requirements including all applicable system interface requirements as described in the standard for the entire duration of specified useful life. The qualification pursuant to this specification must be carried out for each type and diameter. The qualification may not be extended following a change in diameter, type of pipe engagement or connection system.

1.6 Test Planning

A test plan (DVP&R) shall be developed to verify and validate all required functions specified in this standard. The responsible entity shall provide any additional testing that's deemed necessary to meet functional requirements; the tables at the end of this document provide summary of all the requirements, and shall be used to develop the test planning.

Samples utilized for Design Verification (DV) shall meet specified drawings tolerance range. Samples for Production Validation (PV) shall meet statistical capabilities for machine and manufacturing processes.

Where necessary, each line item of the test plan (DVP&R) must be supported by a comprehensive test or laboratory procedure. A test plan (DVP&R) agreement by FCA does not relieve the responsible entity from ensuring that the component or system meets all environmental, physical, electrical, mechanical, functional, regulatory and durability requirements including all applicable system interface requirements, in the entity's area of responsibility, as described in the standard for the specified useful life.

1.7 Suggested, Recommended or Specified Components and/or Suppliers

Suggested, recommended or specified components and/or suppliers, if any, are still required to comply to the specification and do not relieve the responsible entity from ensuring that the component or system

On December 01, 2023 0:23 AM (UTC+10) - Userid: T7234G0
PRS_STLA_CHINA@stellantis.com

utilizing these suggested, recommended or specified components and/or suppliers meets all environmental, physical, electrical, mechanical, functional, regulatory and durability requirements including all applicable system interface requirements, in the entity's area of responsibility, as described in the standard for the specified useful life.

1.8 Auxiliary device (Secondary Latch) required characteristics:

a) Two distinct and sequential operations are necessary to certify correct assembly.
b) The first operation must already guarantee that the tests required by the specification were passed.
c) The second operation is carried out only if the first was carried out correctly.
d) If the second operation was already carried out by mistake, the insertion of the interface must be prohibited.

Note : The second operation can be manual handling or automatic

Repeat the following tests with the secondary device inserted:
- 6.3.6 Elevated Temperature Burst Test
- 7.2 Assembly Effort
- 7.3 Pull-Apart Effort
- 7.4 Side load

NOTE: The application of these devices is mandatory for new designs.

2 REFERENCES

Table 1 - References

Document Number	Shield/Designator (if applicable)	Document Title	Downloadable for suppliers from beSTandard
7.N0007		FINISH COMPONENTS VEHICLE INTERNAL/EXTERNAL PROTECTIONS NOISE AT THE BENCH	Y
CS-9003		SUPPLIER REQUIREMENTS FOR VEHICLE AND SERVICE PARTS: RESTRICTED AND PROHIBITED SUBSTANCES	Y
DS-107		BSR DESIGN GUIDELINES	N
FMVSS 301		FUEL SYSTEM INTEGRITY	N
MS-10756		FUEL - ULTRA LOW SULFUR DIESEL - EMISSIONS CERTIFICATION TEST FUEL	Y
MS-10781		FUEL- EU TYPICAL CETANE, ULTRA-LOW SULFUR DIESEL VALIDATION FUEL	Y
MS-11182		FUEL - DIESEL - ULTRA LOW SULFUR CONTAINING 20% BIODIESEL - TEST FUEL	Y
MS-12506		FUEL- E15 TEST FUEL	Y
MS-8004		TIER 2 INDOLENE - EMISSIONS CERTIFICATION TEST FUEL	Y
PF.90197	<S,E>	FUEL BUNDLE AND FUEL SYSTEM JUMPERS - PLASTIC, RUBBER AND METALLIC	Y
PF-EMISSIONS	<E>	IDENTIFICATION OF EMISSIONS ITEMS	Y
PF-SAFETY	<S>	PRODUCT SAFETY - USE OF SAFETY SHIELDS	Y
PS-11346		WARRANTY RETURNED PARTS TESTING AND ANALYSIS PROCEDURES	Y
PS-7300	<D>	PRODUCT QUALITY - USE OF DIAMONDS	Y
QR.00001		GLOBAL PRODUCT ASSURANCE TESTING (GPAT)	Y
QR-10012		DIMENSIONAL QUALITY REQUIREMENTS	Y
SAE J1645		FUEL SYSTEMS AND COMPONENTS—ELECTROSTATIC CHARGE MITIGATION	N
SAE J2044		QUICK CONNECT COUPLING SPECIFICATION FOR LIQUID FUEL AND VAPOR/EMISSIONS SYSTEMS	N
SAE J2260		NONMETALLIC FUEL SYSTEM TUBING WITH ONE OR MORE	N

On December 01, 2023 02:23 AM (UTC+1) user: T7284CD
PRS_SHLA_CHINA@stellantis.com

Table 1 - References

Document Number	Shield/Designator (if applicable)	Document Title	Downloadable for suppliers from beSTandard
		LAYERS	
SAE J2334		LABORATORY CYCLIC CORROSION TEST	N
SAE J400		TEST FOR CHIP RESISTANCE OF SURFACE COATINGS	N
SD-11597		FUEL COMPATIBILITY LIST	Y

3 DEFINITIONS/ABBREVIATIONS/ACRONYMS/SYMBOLS

BICEEPR&U	Body, Interior, Chassis, Electrical, Engine System, Powertrain, Restraints & UConnect
BSR	Buzz, Squeak and Rattles
CAD	Computer Aided Design
Continuing Conformance (Pre-Capability)	Continuing conformance tests are used to demonstrate process compliance and must be completed using production castings, tooling, and processes. Continuing conformance tests are to be done at the specified frequency, until process capability or confidence is demonstrated per the defined test criteria. Continuing conformance tests are to be conducted at the specified frequency, for a minimum of three production days, or nine production runs, whichever is more stringent.
Continuing Conformance (Post-Capability)	Post capability continuing conformance test program may be implemented after successful completion of the continuing conformance requirements. Tests must be completed with production parts on a continuing basis. Samples for these tests must be selected on a random basis to represent the entire production population as much as possible. In the event that any portion of these tests is not met, the reaction plan specified in the Supplier control plan shall be invoked.
CMVSS	Canadian Motor Vehicle Safety Standards
DV	Design Validation
DVP&R	Design Verification Plan & Report
DFMEA	Design Failure Mode and Effect Analysis
ECE	Economic Commission for Europe
EM	Electro-Mechanical
FCA	Fiat Chrysler Automobiles
FCA US LLC	Used to be Chrysler Group LLC
FCA Italy S.p.A.	Used to be Fiat Group Automobiles S.p.A.
FMVSS	Federal Motor Vehicle Safety Standards
LOT	A group of couplings that can be traced to a single assembly set-up or material lot. No more than one week production in a lot.
NVH	Noise, Vibration and Harshness
PV	Product Validation
PPAP	Production Part Approval Process
PSI	Integrated Test Plan (Piano Sperimentale Integrato)
QRE	Quality & Reliability Engineer/ Engineering
TFO	Fiat Design and Organization Procedures
Unexposed coupling	One that has not been used or deteriorated since manufacture.

on December 01, 2023 02:26 AM (UTC+1) - user id: T7284C0
PRC STELLANTIS CHINA@stellantis.com

4 SPECIAL TEST EQUIPMENT

Not applicable.

4.1 Test environment (unless otherwise specified)

Test temperature shall be maintained at 23 ± 5°C.
Atmospheric pressure shall be 860 mbar +/- 40 mbar.
Relative Humidity shall be at 45 to 70%.

5 ENVIRONMENTAL REQUIREMENTS

Quick Connectors shall be designed for continuous use:
Operating pressure up to 6 bar.
Operating vacuum down to -0.5 bar.
Operating temperature -40°C to 120°C and short term exposure 135°C for 20 min**.**

NOTE: New connector designs using the same materials as previously tested connectors may use the original results as surrogate data for 5.1 Corrosion, 5.2 Zinc Chloride Resistance, 5.3 Calcium Chloride Resistance, 5.4 Chemical Exposure, 5.5 Internal Fuel Resistance Test**.**

5.1 Corrosion

Purpose:
The fuel system quick connect fitting shall remain functional for the useful life of the vehicle, 15 years /150,000 miles when exposed to the corrosive environment.
If there are no metal components in the quick connector, corrosion does not need to be tested.

Procedure:
Corrosion requirement and vehicle level testing per **CS.00081** for 15 year functional life. Laboratory test is per **SAE J2334** for 180 cycles as called out in **CS.00081** using the immersion method.

Acceptance Criteria:
At completion of testing, the quick connector shall be free of cracks and shall meet the functional requirements of Sections 6.1 Appearance, 6.3.5 Dynamic Impact, 7.1 Leak, 7.2 Assembly Effort, and 7.3 Pull-Apart Effort.

5.2 Zinc Chloride Resistance

Purpose:
The fuel system quick connect fitting shall remain functional for the useful life of the vehicle, 15 years /150,000 miles when exposed to the Zinc Chloride environment.

Procedure:
See **SAE J2044** Section 7.2 Zinc Chloride Resistance.

Acceptance Criteria:
At completion of testing, the quick connector shall be free of cracks and shall meet the functional requirements of Sections 6.1 Appearance, 7.1 Leak, 7.2 Assembly Effort and 7.3 Pull-Apart Effort.

on December 01, 2023 02:23 AM (UTC+1) - userid: T7284G0
PRS_STLA_CHINA@stellantis.com

5.3 Calcium chloride resistance (EMEA Only)

Purpose:

The fuel system quick connect fitting shall remain functional for the useful life of the vehicle, 15 years /150,000 miles when exposed to the Calcium Chloride environment.

Procedure:

A. Insert the female connector into a male connector. Insert a nylon line onto the barb end of the quick connector. Plug the male connector and nylon line to prevent chemical aggression inside the coupling.
B. Submerge the assembly in 50% aqueous calcium chloride bath for 200 h at a temperature of 60°C. The concentration of the solution may change during the test.
C. Remove the couplings without rinsing or cleaning them. Expose the couplings to hot air at 60°C for 24 h.
D. Clean connection with warm water, wipe dry.
E. Inspect the samples and make sure there are no cracks, deformations, flaking or other damage. No red rust spots are permitted on any metal components.

Acceptance Criteria:
At completion of testing, the quick connector shall be free of cracks and shall meet the functional requirements of Sections 6.1 Appearance, 7.1 Leak, 7.2 Assembly Effort and 7.3 Pull-Apart Effort.

5.4 Chemical Exposure

Purpose:
The quick connect shall be resistant to degradation from automotive chemicals for the useful life of the vehicle, 15 years / 150,000 miles.

Procedure:
The quick connect shall be subjected to the specific chemicals and conditions per **SAE J2044** Section 7.3 External Chemical and Environmental Resistance, additionally test (80% FAM DIN B + 10% methanol + 10% ethanol + 1cc/l H2SO4), windscreen washer liquid, and urea to the same conditions.

Additionally apply 40 ± 2% sulfuric acid (sg. = 1.3) to polypropylene non-woven felt pad and manually rub for 10s the outer surface of the most stressed areas of the connector. After the specified time, check parts under light to look for failure. There must not be any sign of degradation, such as cracks or melted resins, among others.

Acceptance Criteria:
At completion of testing, the quick connector shall be free of cracks and shall meet the functional requirements of Sections 6.1 Appearance, 7.1 Leak, 7.2 Assembly Effort and 7.3 Pull-Apart Effort.

5.5 Internal Fuel Resistance Test

Purpose:
The quick connect assembly shall be capable of functioning after being exposed to all commercially available domestic and foreign fuel blends. Testing in this section is a minimum recommended testing. Supplier is responsible for the part performing after exposure to any of the fuels in **SD-11597** or any fuels in the markets where the product will be sold.

The quick connect assembly must meet for the useful life of the vehicle, 15 years / 150,000 miles when exposed to any fuel that may be used on the vehicle.

on December 01, 2023 02:23 AM (UTC+1) - Userid: T7284G0
PRS_STLA_CHINA@stellantis.com

The materials used in a gasoline/diesel/flex fuel assembly must function with continuous exposure to liquid fuel and fuel vapors, including alcohol blends as described in the FCA Fuels Compatibility List in **SD-11597.**

Minimum fuels for gasoline or E10 applications are:

- A5 (E22) per **SD-11597**
- Sour Gas - CARB Phase 3 (Unleaded with 10% ethanol) + 2.6mL/L Butyl Hydroperoxide OR 10.4 mL/L Lauroyl Peroxide
- FamB – 42.25% Toluene + 25.35% Iso-octane + 12.675% Disobutene + 4.22% Ethanol + 15% Methanol + 0.5% deionized water per **DIN 51604**

Minimum fuels for flex fuel applications are:

- B2 (E85) per **SD-11597**
- E100

Minimum fuels for diesel applications are:

- C5 (Biodiesel 100% Rape Seed) per **SD-11597**
- C6 (Biodiesel (Rape Seed) with 20% by volume MS-10781 high cetane diesel fuel per **SD-11597**
- C7 (Biodiesel B-20 **MS-11182** with ultra low sulfur diesel **MS-10756**) per **SD-11597**
- Biodiesel B-30 **MS-11182** with ultra low sulfur diesel **MS-10756**

Procedure:
For gasoline and fuel diesel, temperature is 90 °C (194 °F).
For gasoline temperature is 60°C (140°F) only North America Market.
Fuel shall be circulated during soak. External air temperature is 120 °C (248 °F) for 250 hours; then, 130 °C (266 °F) for 12 hours. Repeat the external air temperature cycle a total of four times (1048 hours total test time), refer to **Figure 3 – Internal Fuel Resistance**.

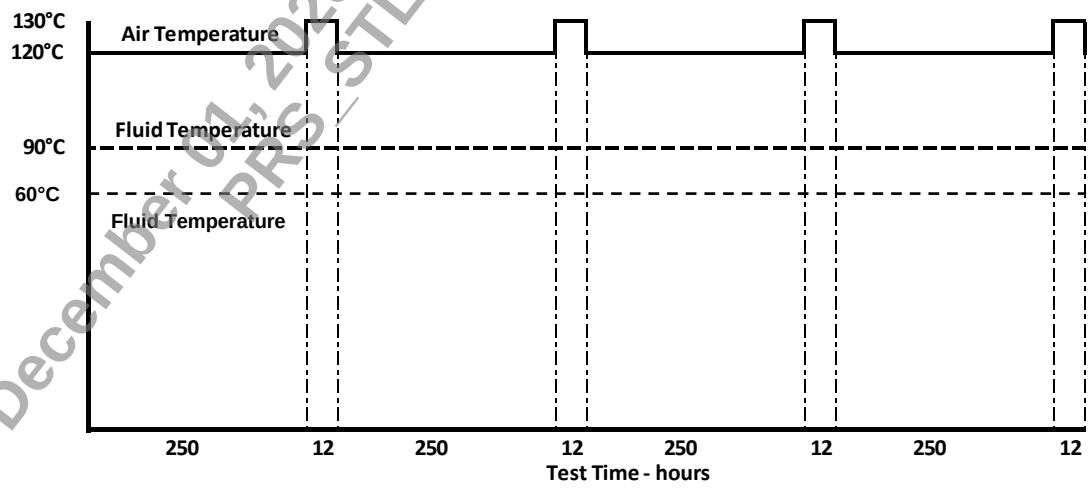

Figure 3 - Internal Fuel Resistance

Acceptance Criteria:
At completion of testing, the quick connector shall be free of cracks and shall meet the functional requirements of Sections 6.1 Appearance, 7.1 Leak, 7.2 Assembly Effort and 7.3 Pull-Apart Effort.

on December 01, 2023 09:23 AM (UTC+1) - useid: T728450
PRS_STLA_CHINA@stellantis.com

5.6 Storage

Supplier must ensure quick connect can be stored for 1 year with no defects to quick connect.

6 APPEARANCE / PHYSICAL / MECHANICAL / ELECTRICAL REQUIREMENTS

No special tools shall be required for installation or removal of the quick connector fittings during production assembly or service repairs. Quick Connect fitting must be compatible with tube end treatment as contained in SAE J2044. Quick Connect fitting must use a dual captured o-ring system. During removal of fitting, o-ring must remain intact and located. Quick Connect fittings must use a redundant latch design. Redundant latch requires the quick connector to be inserted and engaged to the bead form prior to being able to engage the clasp.

During insertion of the tube and quick connect, if inserted at any slant from axial direction, o-rings must maintain sealing function without damage to the o-ring from the tube end. Internal design of Quick Connect fitting must not cause additional restriction in the fuel supply or return system. All external dimensions must be approved by FCA Engineering.

The O-rings may have been previously lubricated by the supplier with non-silicone oils, for which the supplier has guaranteed compatibility with the engine control and fuel process (lambda probe). Therefore, the specification of the lubricant must be sent to FCA Materials Laboratories for approval, also indicating the maximum quantity used.

The insertion of male connector either correct or incorrect (e.g. not coaxial) must not damage the O-ring or male connector itself, compromising the seal after the operation is correctly repeated.

Tools for connection on the vehicle assembly line (either automatic or semiautomatic, or manual) must be approved beforehand by the supplier of the pipe/component/assembly where the quick coupling is located.

The connector can be disconnected and reconnected up to 30 times during the lifespan of the vehicle.

6.1 Appearance Requirements

Plastic parts
The parts must be free of cracks, dents, breakage, sharp parts, burring, streaks, pulls, bleeding of any "loads/reinforcements", voids or any other defect that may compromise aesthetics, operation or assembly. The coupling areas for any inserts must not present discontinuity in adherence, pulls, distortions, incompleteness, etc.

If any parts are joined by welding (hot blade, ultrasound, etc.), the appearance/hold of the welding must be continuous and homogenous, the union seam must not have any interruptions and have a homogenous appearance.

The overflow material must not be degraded (detected by a carbonaceous appearance). The forced stress on the welding zones (e.g. destructive control) may cause the elements to separate; in any case, the welding seam or spot must be homogenous and cohesive on both surfaces.

Metal parts
All parts must be free of cracks, dents, breakage, internal or external oxidation, sharp parts, burring, streaks, or other defect that may compromise aesthetic, operation and assembly.

on December 01, 2023 02:23 AM (UTC+8) - user: TZ284G0
PRS_STLA_CHINA@stellantis.com

The anticorrosion treatments/processes must be uniform with the thickness indicated on the drawing or on the self-qualification document.

6.2 Physical Requirements

Supplier shall demonstrate dimensional capability for all key dimensional characteristics, identified on CAD files, during DV and PV phases including physical interface requirements e.g.: Fastening points; Interface constraints; Interface materials etc.

6.2.1 Material Requirements

Supplier shall review all materials used in the quick connectors with FCA materials engineering.

6.2.1.1 Absence of harmful or potentially harmful materials in processes

The quick connectors shall not contain materials that are considered harmful to people or the environment. The quick connectors must meet the requirements of **CS-9003**.

6.2.1.2 Melting Temperature

Differential Scanning Calorimetry (DSC) Analysis shall be included as part of PV testing and results retained for usage as a master sample.

6.2.1.3 O-Ring Material (For Reference)

If supplier would like to use a different type of O-ring, new material must be discussed with Stellantis engineering for additional validation and approval.

APAC - EMEA - NA Market	
O-rings in contact with fuel	External safety and antmosferic contact O-ring
FKM "B" - DOWTY 9798 - PARKER - V 3642 - 75 - LE J.F. - 6 DF 1882 (DOWTY 9815 A)	FVMQ (FLUOROSILICON) - DOWTY/PARKER (DOW CORNING) 28075 - DOWTY 8975 - PARKER L 3355-70 - LE J.F. - 7 SF 2081
LATAM Market (BRASILIAN Ethanol)	
O-rings in contact with fuel	External safety and antmosferic contact O-ring
FKM "B" - DOWTY 9798 - PARKER - V 3642 - 75 - LE J.F. - 6 DF 1882 (DOWTY 9815 A)	FKM - TR 10 - -29°C ("GLT") - DOWTY 9787 - PARKER - V 8592 - 75 - LE J.F. - 7 DF 2088

on December 01, 2023 02:23 AM (UTC+1) - userid: T7284G0
PRS_STLA_CHINA@stellantis.com

6.2.2 Dimensional Requirements

The Quick Connect shall meet the **dimensional requirements specified in the released engineering drawing** and shall adhere to the procedures and guidelines identified in QR-10012, Dimensional Quality Requirements.

6.2.3 Composition, dimensions and tolerances

According to the drawing and/or Annexes of this document.

In the case of application with a cartridge on the component, the dimensions and tolerances must be agreed upon with the various suppliers.

6.2.4 Internal Cleanliness Requirement

Purpose:
The quick connectors shall not have contaminants that affect the vehicles performance for the useful life of the vehicle, 15 years / 150,000 miles.

All openings shall be protected against damage, dust, dirt, and / or other contamination to the connectors during shipping and storage.

Procedure:
A. Testing devices must be utilized which can safely and accurately flush the interior of the quick connector with solvent, effectively remove any contaminants and foreign material from the interior surface, and accurately measure that material and its weight.
B. Test can be conducted per **SAE J2045** or **7.G0200**.
C. Repeat quick connector flushing steps until it is verified that >90% of contaminants are removed by repeating the procedure until measured value is <10% of first flush.

Acceptance Criteria:
- Total contamination shall not exceed 0.75 mg/dm^2.

6.2.5 Indications on drawings

The following requirements must be indicated on the drawing:
- The number of specification
- The dimensional class of the coupling based on the diameter/form of the male
- The type of male
- The material of the O-rings according to specification
- The materials used in the body and in the retainer
- The weight
- The conductibility

6.3 Mechanical Requirements

6.3.1 Misalignment Assembly Test

During a manual assembly process, all possible assembly angle of Quick Connect coupling shall not damage (dislodge, cut, lacerate) all sealing O-rings. Use a suitable tensile tester. Use a metallic tester spigot with minimum diameter (refer to **SAE J2044** – Test pin table). No additional coupling lubrication is permitted.

On December 01, 2025 10:23 AM (UTC+1) - userid: T7284G0 PRS_STLA_CHINA@stellantis.com

Definitions:

A. Spigot: Male Quick connect;
B. F_V: plug-in vertical force tracked during the test;
C. M_V: reaction torque to the rotation axis, it is a torque applied to the fixture frame system's rotation axis and it shall be 2,5 Nm (±0,5 Nm) (example of possible torque application, refer to **Figure 6**. $M_V \cong F \cdot l = \mu mg \cdot l$
D. αi: test assembly angle, its value shall be between 5° and αMAX - 5° with a step not higher than 5°;
E. αMAX: maximum test assembly angle; during the test 6.3.8 with a correct set-up, it is "the minimum applicable angle for failing the rotation system because the spigot does not slide to the correct final position and the ;
F. MVP: Maximum Value to Perform = αMAX - 5°

Procedure:

A. Attach the spigot on the moving crosshead's jaw in vertical position;
B. Attach the Quick Connector body stem W/O latch to the fixture with following requirements (**Figure 4**)
   a. Apply α angle to QC body
   b. Insert the spigot as far as it takes contact with QC internal wall ("Contact Point")
   c. Lock the QC body position in order to match the Rotation Center on Rotation axis of the fixture frame system;
   d. Pull-down the spigot to the final assembly position in order to check the correct rotation of the fixture system.
C. Rise-up the spigot 30 mm above the contact point (**Figure 5**)
D. Stop Quick Connector body at αi according to its definition (or different single/matrix α target agreed with FCA PD Team)
E. Apply a compression load, at a rate of 51 mm/min ±5 mm/min (2 in/min ±0,2 in/min) as far as the spigot gains the final connected position;
F. Repeat 5 times from Step C to Step E for each αi;
G. Report all items below:
   a. QC O-ring status (picture and textual description);
   b. F_V max value reached during the test;
H. Repeat the test for 30 samples.

Example:

αMAX = 18°

MVP = αMAX -5° = 13°

Table 2 - Misalignment Assembly Test

αi	Standard Values	Values to Perform
α1	5°	>5°
α2	10°	>10°
α3	15°	>13°
α4	20°	-
α5	25°	-
α6	30°	-

on December 01, 2023 9:23 AM (UTC+1) - uuid: T7284G0
PRS_STLA_CHINA@stellantis.com

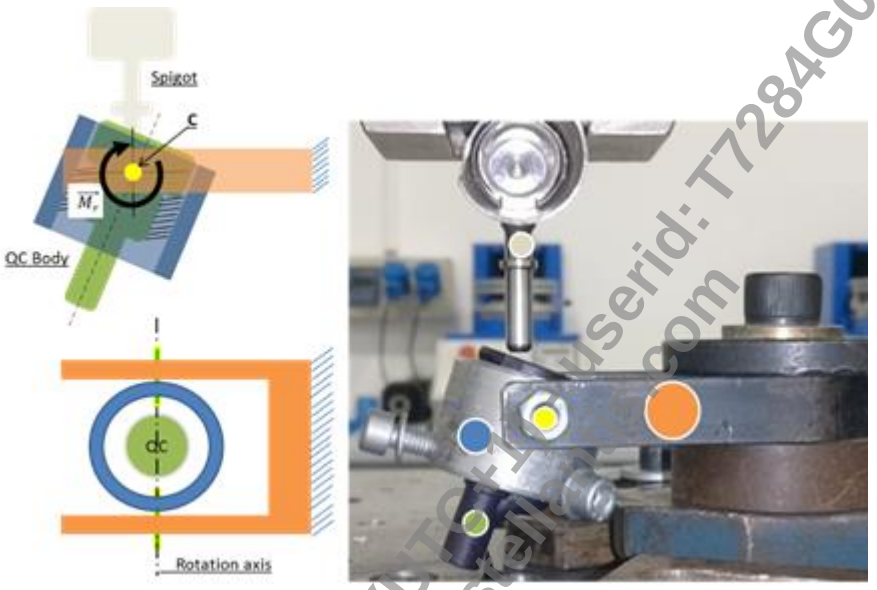

Figure 4 - Fixture Frame System

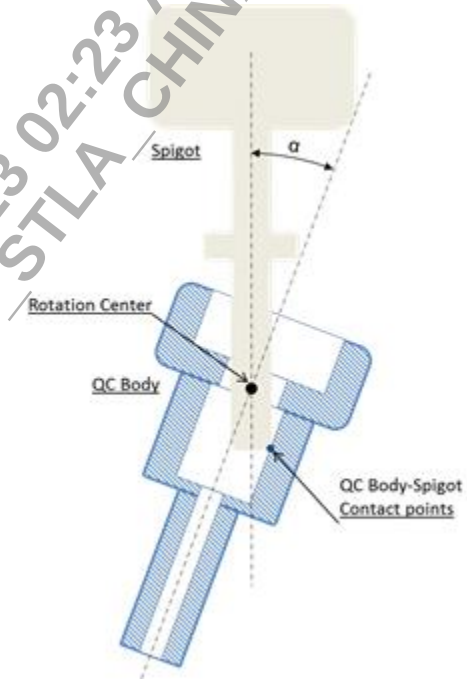

Figure 5 - Contact point

on December 01, 2023 02:23 AM (UTC+1) userid: T7284G0
PRS_STLA_CHINA@stellantis.com

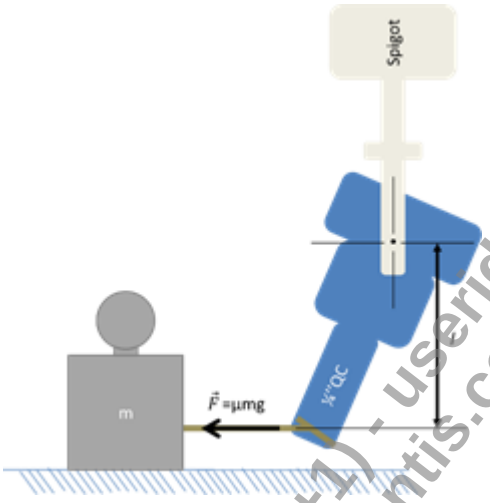

Figure 6 - $M_y \approx F \cdot l = \mu mg \cdot l$

Alternative Procedure:

A. Test quick connect as is with no additional lubrication on part or mating component.

B. Attach quick connect to test fixture. Identify the orientation of the quick connect based on what is 12, 3, 6 and 9 o'clock positions.

C. Wipe test pin to prevent accumulation of lubrication

D. Insert assembly test pin into the quick connect at a rate of 51 mm/min ± 5 mm/min(2 in/min ± 0.2 in/min) at the maximum angle the housing allows in the 12 O'clock position

E. Remove the test pin and inspect the quick connect for any damage or displaced internal components.

F. Repeat this in each position identified in step B twice, including a straight insertion twice.

Acceptance Criteria:

- No damage to end form components.
- No displacement of end form components.
- The quick connector shall meet the functional requirements of Section 7.1 Leak Test <S,E>.

6.3.2 Buzz, Squeaks and Rattle

Anti-rattle devices shall be employed where necessary to prevent all BSR. This shall be specifically documented during mechanical vibration testing at the component level. Noise level shall be evaluated with the component assemblies mounted in the appropriate vehicle location.

The BSR requirements shall be checked per **7.N0007** "FINISH COMPONENTS VEHICLE INTERNAL/EXTERNAL PROTECTIONS or DS-107 "BSR DESIGN GUIDELINES".

On December 01, 2023 02:23 AM (UTC+1) - userid: T7284G0
PRS_STLA_CHINA@stellantis.com

6.3.3 Part Identification and Marking

Not Applicable.

6.3.4 Traceability

Not Applicable

6.3.5 Dynamic Impact

Purpose:
The Quick Connector must be able to withstand a moderate impact without damage or leakage.

Procedure:
A. Pre-cut 6" of specified fuel line.
B. Pre-cut 6" of nylon that has the specific quick connect to be tested pre-assembled.
C. Mount the fuel line - quick connect assembly into a Charpy Impact Testing Machine such that the quick connect will be impacted. Refer to **Figure 7 – Dynamic Impact Test Setup 180° and Figure 8 – Dynamic Impact Test Setup 90°**.
NOTE: The fuel line must be rigidly mounted.
D. Prepare the impact hammer such that the surface will impact the whole length of the quick connect.
E. Impact the quick connect system with 10 Joules of impact energy at -30°C
F. Leak check the quick connect per section 5.1
G. If the system endures the 10 Joules of impact energy with no leakage, repeat the test at 20 Joules using a new sample. Continue increasing the impact energy in increments of 10J until separation/leakage of the quick connect occurs.

NOTES:
- For plastic QC's perform test at the 12, 3, 6 and 9 o'clock positions.
- For plastic QC's test to be performed using plastic & steel male endforms
- For metallic QC's test to be performed using steel male endform

Acceptance Criteria:
The quick connect assembly shall meet Section 7.1 Leak Test <S,E> at the minimum impact energy shown below.
Metallic: 50J
Hybrid: 30J
Plastic: 10J

on December 01, 2023 02:23 AM (UTC+1) - userid: T7284G0
PRS_STLA_CHINA@stellantis.com

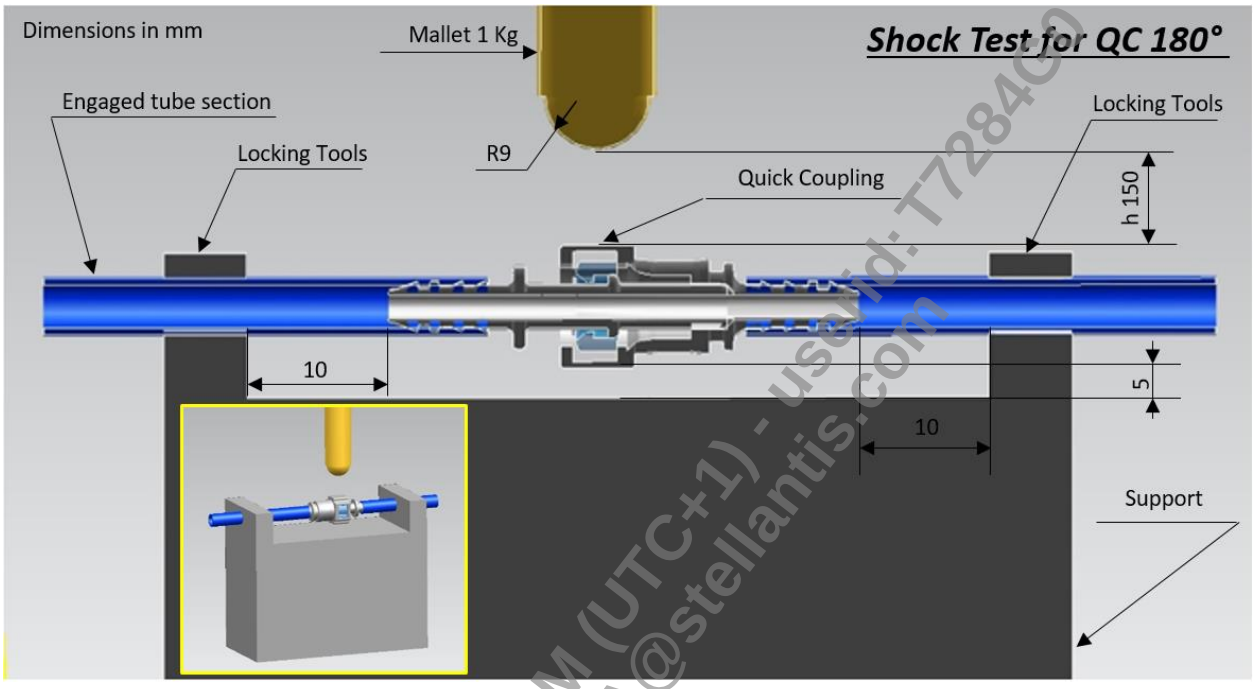

Figure 7 - Dynamic Impact Test Setup 180° Quick Connect

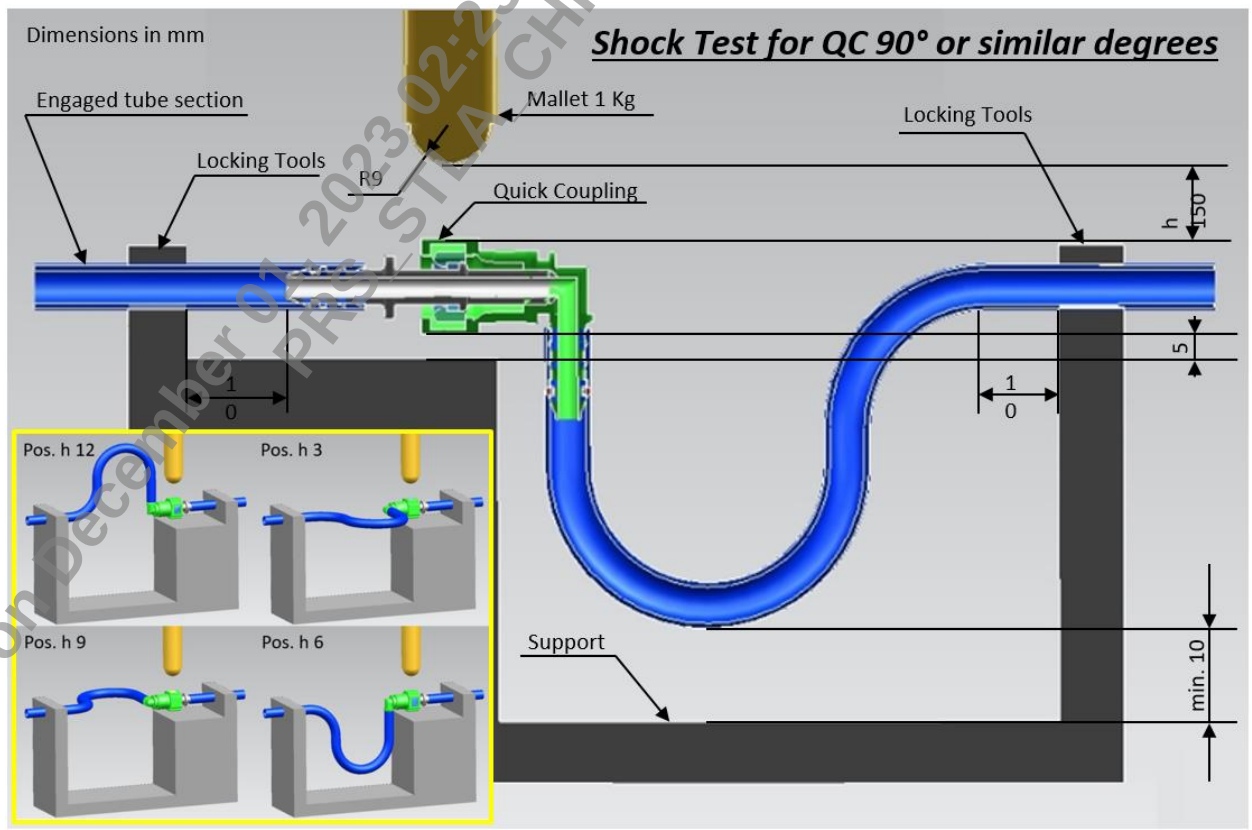

Figure 8 - Dynamic Impact Test Setup 90° Quick Connect

On December 01, 2023 02:23 AM (UTC+1) - user id: T728469
PRS_STELLA_CHINA@stellantis.com

6.3.6 Elevated Temperature Burst Test

Purpose:
The fuel system quick connect fitting must remain connected to the fuel line for the useful life of the vehicle, 15 years / 150,000 miles.

Procedure:
See **SAE J2044** Section 7.6 Elevated temperature Burst.

Acceptance Criteria:
A. Minimum burst pressure for liquid fuel quick connect couplings 3447 kPa (500 psig).
B. Minimum burst pressure for vapor/emission quick connect couplings 689 kPa (100 psig).

NOTE: The burst test is for the quick connector only. Leakage or rupture of the hose is not a failure. If the hose fails below the minimum requirement, the test must be rerun with hose capable of higher pressure, if practical.

6.4 Electrical Requirements

6.4.1 Electrical Resistance (ESD) Requirements <S>

Purpose:
The quick connect fitting must provide an adequate ground path for the intended useful life of the vehicle.

Procedure:
Per **SAE J1645**, verify the quick connect fitting meets grounding requirements between the fuel line assembly and the ground point. The inside surface of the entire quick connect fitting is to be the ground path.

Per **SAE J1645** measure the D.C. resistance using an ohmmeter from one extreme end of the fuel systems assembly to the other end for each of the fuel lines involved. If at all possible, measurements of the assembly attached to the vehicle should also be done.

Acceptance Criteria:
D.C. resistance between any metal or conductive fuel system component in direct contact with the liquid fuel during vehicle operation shall not exceed 1 000 000 Ω measured using a 500-V source.

7 FUNCTIONAL REQUIREMENTS

This section defines the minimum functional requirements for quick connector couplings in fuel tubing systems.

NOTE:
The qualification pursuant to this specification must be carried out for each type and diameter. The qualification may not be extended following a change in diameter, type of pipe engagement or connection system

7.1 Leak Test<S,E>

Purpose:
The fuel system quick connect fitting shall maintain liquid and vapor integrity for the intended useful life of the vehicle.

on December 01, 2025 02:23 AM (UTC+11) - Userid: T7284G0
PRS_SEA_CHINA@stellantis.com

- High Pressure Leak Test is required only for quick connectors used in liquid fuel applications.
- Low Pressure Leak Test is required for quick connectors used all applications.
- Vacuum Leak Test is required only for quick connectors used in vapor applications.

Equipment and method used to accomplish these tests must be approved by FCA Engineering.

7.1.1 Low Pressure Leak Test <S,E>

Procedure:
Test per **SAE J2044** Section 6.1.1 Low Pressure Leak Test procedure.

Acceptance Criteria:
The maximum leak rate shall be no more than 0.5 cc/min. of air (or equivalent) leakage.

7.1.2 High Pressure Leak Test <S,E>

Procedure:
Test per **SAE J2044** Section 6.1.3 High Pressure Test procedure.

Acceptance Criteria:
The maximum leak rate shall be no more than 0.5 cc/min. of air (or equivalent) leakage.

7.1.3 Vacuum Leak Test

Procedure:
Test per SAE J2044 Section 6.1.5 Vacuum Leak Test procedure.

Acceptance Criteria:
The maximum leak rate shall be no more than 0.5 cc/min. of air (or equivalent) leakage.

7.1.4 In-Process Leak Test

SAE J2044 6.1 Leak Test (In-Process) Also tests between dual O-ring systems

7.2 Assembly Effort<D>

Purpose:
The fuel system quick connect assembly effort shall meet reasonable ergonomic efforts to fully assemble (latch or retain) the mating tube end into the connector for the useful life of the vehicle.

Procedure:
See **SAE J2044** Section 6.2 - Assembly Effort test procedure.

Acceptance Criteria:
The maximum assembly force is 67 N (15 lb.) for sizes < 11mm male tubes and 111N (25 lb.) for sizes ≥ 11mm male tubes for new fittings and 111 N (25 lb.) for sizes < 11mm male tubes and 156 N (35 lb.) for sizes ≥ 11mm male tunes for aged fittings.

7.2.1 Assembly Effort for Integrated Female (Cartridge) Fitting

Force of inserting the integrated female (cartridge) on the component:

On December 01, 2023 02:23 AM (UTC+1) - userid: T128460
PRS_STLA_CHINA@stellantis.com

Dry the O-rings on the cartridge for 24 hours at 120 ± 2°C.
Then lubricate them lightly with mineral based oil (5cst).
Insert the cartridge at a speed of 50±5 mm/min along the longitudinal axis of the housing seat into the housing and use a suitable tool to measure the insertion force.
The maximum insertion force must not exceed 650 N.

7.3 Pull-Apart Effort<D>

Purpose:
The fuel system quick connect fitting must remain connected to the fuel line for the useful life of the vehicle.

Procedure:
See **SAE J2044** Section 6.3 - Pull-Apart Effort test procedure.

Acceptance Criteria:
For liquid fuel quick connectors that have not been exposed to fuel or chemicals, the force required should be 450N minimum.

For liquid fuel quick connectors that have been exposed to fuel or chemicals, the force required should be 297N minimum.

For vapor quick connectors that have not been exposed to fuel or chemicals, the force required should be 222N minimum.

For vapor quick connectors that have been exposed to fuel or chemicals, the force required should be 75N minimum.

7.4 Side Load Capability

Purpose:
The fuel system quick connect fitting must be able to withstand side loads typical of what might be imposed by hose routing in a vehicle application as well as from having the hose pushed aside to reach other objects on the vehicle during service procedures for the useful life of the vehicle.

Procedure:
See **SAE J2044** Section 6.4.1 - Side Load Capability test procedure.

Acceptance Criteria:
See **SAE J2044** Section 6.4.2 and 6.4.3 - Acceptance Criteria.

7.4.1 Side Load Mechanical Resistance for an Integrated Female (Cartridge)

Following the test specified below, depressurize and increase the side load to detachment/breakage. The coupling must not break or show signs of fracture before reaching the load specified below.

on December 01, 2023 02:23 AM (UTC+1) - userid: T7284G0
PRS_GTLA_CHINA@stellantis.com

Outer diameter of male connector (inches)	Side load (N) Straight cartridge	45°cartridge
1/4	200	
5/16	225	
3/8	311	
7/16 and above	400	
Outer diameter of male connector (mm)	**Side load (N)**	**Side load (N)**
8	225	336
10	310	
12 and above	400	

Collapse of the female connector is permitted.

Insert 500 mm of the specified pipe on the female connector. Fix the coupling to the equipment described below in **Figure 9 - Side Load Capability for an Integrated Female (Cartridge)**, making sure the pipes are perfectly connected to the test equipment and that the equipment is correctly anchored.

Pressurize with air to 10.34 ± 0.35 bar (1034 kPa ± 35 kPa) for fuel couplings and 0.7 ± 0.02 bar (70 ± 2kPa) for vapor couplings. Pull on the pipe, in the direction perpendicular to the female nozzle considered most critical, at a speed of 13 ± 5 mm/minute until reaching the load specified in the table.

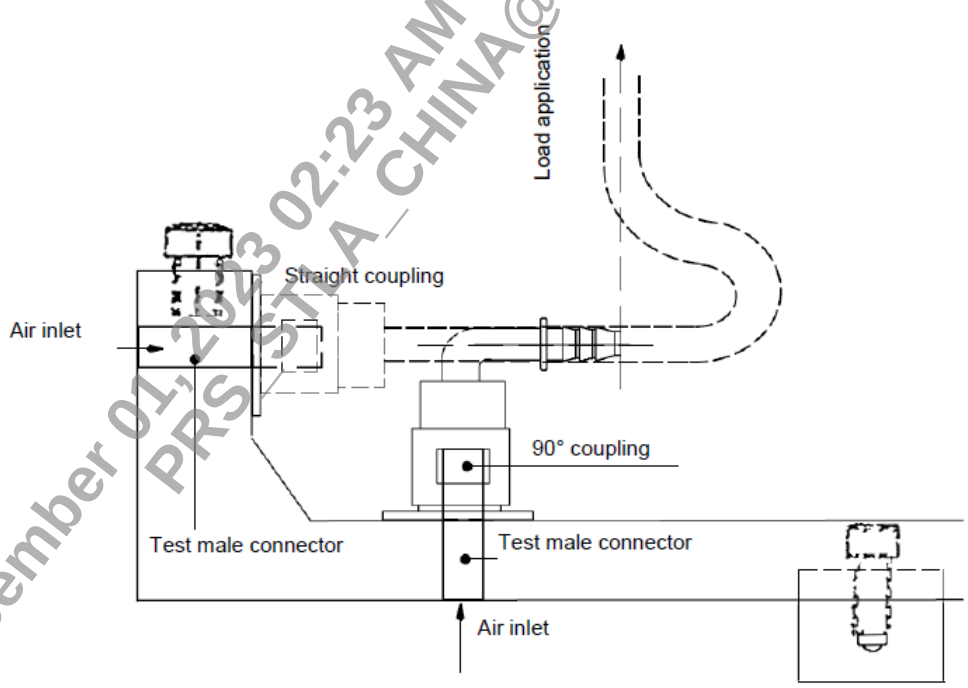

Figure 9 - Side Load Capability for an Integrated Female (Cartridge)

7.5 Flow Restriction

Purpose:
Quick connect couplings shall be designed to provide minimal flow restriction for the useful life of the vehicle.

on December 01, 2023 02:23 AM (UTC+8) - userid: T7284G0
PRS_STLA_CHINA@stellantis.com

Procedure:

A. Insert quick connect into its intended tubing
B. Connect quick connect and tube assembly to pressure source
C. Using Stoddard or equivalent density fluid for fuel or air for vapour, measure the pressure drop across the assembly with a flow from 0-200 liter/minute in increments of 10 liters/min.
D. Perform the flow in both directions; male to female and female to male.

Acceptance Criteria:
The load losses measured with flow rates of SHELL SOL TD (Tab. 55999/06) or a fluid with equivalent viscosity and density, at 100 and 200 l/h must comply with what is measured, in the same conditions, with an equivalent hole in a thin wall and specified as follows:

Diameter (inches)	Heading	Equivalent hole (mm)
5/16	with 6 x 8 mm tube	4.5
3/8	with 8 x 10 mm tube	5.5
1/2	with 10 x 12.3 mm tube	7.5
5/8	with 13.5 x 16 mm tube	9.5

7.6 Gravel resistance test

Purpose:
The Quick Connector must be able to withstand impact from stones or other road debris.

Procedure:
Insert on the female and male 100 mm of tube that has been stiffened by inserting a metal rod for 20 mm.
Insert the male on the female and, if the body of the quick coupling is plastic, stabilize the assembly in the cooling cell at -30°C for 6h.
Keeping the sample at -30°C if the body is plastic, or at ambient temperature if metal, carry out the gravel test according to standard **SAE J400** using 4.7L of gravel.

Acceptance Criteria:
At completion of testing, the quick connector shall meet the functional requirements of Section 7.1 Leak Test, 7.2 Pull Apart effort and 7.4 Side Load.

8 SAFETY AND REGULATORY REQUIREMENTS

Supplier shall ensure compliance to all applicable regulatory requirements for the (Add part/commodity name here) in the markets where the vehicles are sold.

8.1 Vehicle Impact

Vehicle's subjected to any of the impact and rollover tests described within **FMVSS301**, must meet the fuel spillage requirements as described within **FMVSS301**. The supplier shall work with FCA to resolve any impact vehicle issues.

on December 01, 2023 02:23 AM (UTC+1) - userid: 57284G0
PRS_STLA_CHINA@stellantis.com

8.2 Hydrocarbon Permeation Emissions<E>

Purpose:
The Fuel System Quick Connect Fittings hydrocarbon contribution to the SHED, during an EPA Diurnal Test Cycle, shall not adversely affect the Vehicle Level permeation requirements.

9 RELIABILITY / DURABILITY REQUIREMENTS

Reliability/durability test procedures are required to demonstrate at engineering level (with a given probability and level of confidence) that a system or component will perform its intended functions over the expected useful life, under all anticipated operating conditions and environments as specified in preceding sections. The tests must ensure that expected design and process variations are covered, resulting in adequate product robustness. Test to Failure (TTF) is the preferred test methodology.

A. Vehicle Useful or Service Life Requirements: 15 years / 150,000 miles

B. Minimum Reliability and Confidence demonstration: R95 / C90

9.1 System and Vehicle Level Requirement

FCA Engineering shall perform system and vehicle level tests on program phase vehicles (such as, System bench tests, Plywood Buck, Hardware in Loop, Corrosion, PG Driving, BSR, Hot/Cold Trips, etc.) during program development. Examples of FCA Proving Grounds include HMSR, SXV, PT-7, L4S, etc. or reliability validation on 336 vehicle fleets to verify system interface and customer usage applications during different stages of program development (Examples: FDV, VP, PS and Job 1).

Any degradation to functional and/ or performance requirement involving component reliability/ durability issues shall be promptly addressed by component Supplier. Corrective actions shall focus on issue prevention and may involve changes to design including any component validation testing.

9.2 Design Failure Modes and Effects Analysis (DFMEA)

The DFMEA must address and correlate to all of the functions which are specified in this Performance Standard. All potential Failure Modes, as described, must be evaluated and taken into account in the DFMEA risk analysis. FMEA analysis shall be included in timing plan shared and agreed with supplier at project start. A complete DFMEA is normally property of suppliers and available for examination at their site. Supplier will be responsible for DFMEA execution and content accuracy; FCA component or system specialist shall analyze and approve supplier's analysis.

9.3 Latch Fatigue and On/Off Insertion for Serviceability

Purpose:
The fuel system quick connect fitting must remain functional over its design life of 15 years or 241,400 Km (150,000 miles) with exposure to On/Off insertion for a demonstrated reliability of R95C90.

Procedure:
1. Following Life Cycle Testing:
   a. Disassemble the quick connect to the fuel line.
   b. Reassemble the quick connect to the fuel line.
2. Repeat disassemble and reassemble for a total of 30 times

on December 01, 2023 02:23 AM (UTC+1) | User: EPS_STLA_CHINA@stellantis.com | Id: T7284G0

Acceptance Criteria:
At completion of testing, the quick connector shall have no physical damage and shall meet the functional requirements of Sections 7.1 Leak Test, 7.2 Assembly Effort and 7.3 Pull-Apart.

9.4 Life Cycling

Purpose:
The fuel system quick connect fitting must remain functional over its design life of 15 years or 241,400 Km (150,000 miles) with exposure to vehicle environment cycles for a demonstrated reliability of R95/C90.

Procedure:
See **SAE J2044** Section 7.5.
At the completion of testing, perform quick connector disassembly, reassembly per section 9.3.

Acceptance Criteria:

At completion of testing, the quick connector shall have no physical damage and shall meet the functional requirements of Sections 7.1 Leak Test, 7.2 Assembly Effort and 7.3 Pull-Apart.

9.5 Rocker

Purpose:
The fuel system quick connect fitting must remain functional over its design life of 15 years or 241,402 Km (150,000 miles) with exposure to vehicle environment for a demonstrated reliability of R90/C90.

Procedure:
A. Pre-cut 12" of a fuel line with and endform.
B. Pre-cut 12" of nylon that has the specific pre-assembled quick connect to be tested.
C. Mount the end-form line into a fuel line clip fixture, refer to **Figure 10 – Rocker test setup**)
D. Assemble the quick connect to the fuel line endform.
E. Mount the free end of the nylon line to the rotational fixture. The free end of the nylon will be the rotating point.
F. Ensure that the locking latch is fastened and towards level ground.
G. Pressurize the assembly to maximum vehicle pressure
   a. MPI: 58 psi
   b. GDI: 74 psi
H. Perform the rocker test by rotating the free end in a 12" stroke in the positive and negative Y direction.
I. One cycle is equivalent to 12 " in the positive "Y" direction or negative "Y" direction.
J. Rotate the quick connect clockwise every 4,000 cycles up to 360,000 cycles for vapor quick connects and 600,000 cycles for liquid quick connects.

Acceptance Criteria:
At completion of testing the quick connector shall be capable of meeting the functional requirements of 7.1 Leak Test and 7.3 Pull-Apart.

On December 01, 2023 02:23 AM (UTC+1) - user id: T7284G0
pbs_STLA_CHINA@stellantis.com

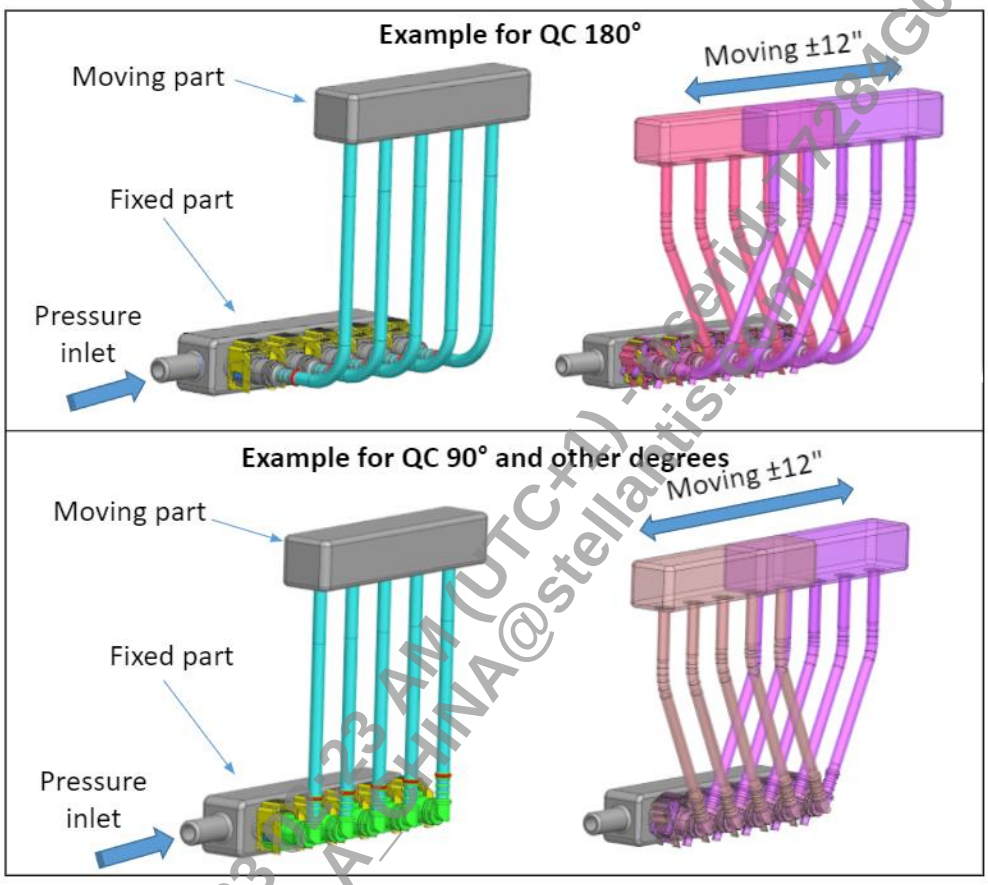

Figure 10 - Rocker test setup

On December 01, 2023 - 23 AM (UTC+1) - serid: T7284G0
PRS_STLA_CHINA@stellantis.com

10 PRODUCT ASSURANCE

The manufacturing facility for a particular component or system shall develop an appropriate PPR (Process Planning Review) and PAT (Process Audit Tool), in accordance with the FCA US PPA (Process Planning and Audit) manual and the FCA Italy program--specific development plan to assure the part meets the quality, durability, and reliability targets throughout the manufacturing process.

10.1 Approved Best Practices

Not applicable.

11 WARRANTY RETURNED PARTS TEST AND ANALYSIS PROCEDURES

The product design, development and manufacturing entities shall comply with **PS-11346**, Warranty Returned Parts and Analysis Procedures.

Annex A
(Normative)
Design Verification & Production Validation

Table A-1 - Design Verification & Product Validation

Function	Section #	Sample Size (Note 1)		Acceptance Criteria		Comments & Responsibility
		DV	PV	DV	PV	
Corrosion	5.1	15	15	P99C90	P99C90	Supplier
Zinc Chloride Resistance	5.2	15	15	P99C90	P99C90	Supplier
Calcium Chloride Resistance	5.3	15	15	P99C90	P99C90	Supplier
Chemical Exposure	5.4	15	15	P99C90	P99C90	Supplier
Internal Fuel Resistance Test	5.5	15	15	P99C90	P99C90	Supplier
Appearance	6.1	100%	100%	All Pass	All Pass	Supplier
DSC	6.2.1.2	N/A	15	N/A	P99C90	Supplier
Internal Cleanliness	6.2.4	15	15	P99C90	P99C90	Supplier
Misalignment Assembly Test	6.3.1	30	30	P99C90	P99C90	Supplier
Dynamic Impact	6.3.5	15	15	P99C90	P99C90	Supplier
Elevated temperature burst	6.3.7	15	15	P99C90	P99C90	Supplier
Electrical Resistance (ESD) Requirement <S>	6.4.1	15	15	P99C90	P99C90	Supplier
Low Pressure Leak Test <S,E>	7.1.1	15	15	P99C90	P99C90	Supplier
High Pressure Leak Test <S,E>	7.1.2	15	15	P99C90	P99C90	Supplier
Vacuum Leak Test	7.1.3	15	15	P99C90	P99C90	Supplier
In Process Leak Test	7.1.4	100%	100%	All Pass	All Pass	Supplier

on December 01, 2023 02:23 AM (UTC+1) - user: IT7284G0
PRS_STLA_CHINA@stellantis.com

Table A-1 - Design Verification & Product Validation						
Function	Section #	Sample Size (Note 1)		Acceptance Criteria		Comments & Responsibility
		DV	PV	DV	PV	
Assembly Effort	7.2	15	15	P99C90	P99C90	Supplier
Pull-Apart Effort	7.3	15	15	P99C90	P99C90	Supplier
Side Load Capability	7.4	15	15	P99C90	P99C90	Supplier
Flow Restriction	7.5	15	15	P99C90	P99C90	Supplier
Gravel Resistance	7.6	15	15	P99C90	P99C90	Supplier
Vehicle Impact	8.1	Vehicle System Test	Vehicle System Test	All Pass	All Pass	FCA
Vehicle Durability	9.1	Vehicle System Test	Vehicle System Test	All Pass	All Pass	FCA
Latch Fatigue	9.3	45	45	R95C90	R95C90	Supplier/ FCA (as applicable)
Life Cycling	9.4	45	45	R95C90	R95C90	Supplier
Rocker	9.6	45	45	R95C90	R95C90	Supplier

NOTE 1: If provided, the sample size shown is the FCA recommended samples per **QR.00001**, which in conjunction with the analysis method indicated best demonstrates the Acceptance Criteria has been met. The supplier is responsible for meeting the Acceptance Criteria specified and may use other valid sample sizes and analysis methods approved by FCA US

NOTE 2: Applicable tests, sample sizes and acceptance criteria shall be per requirements specified in each of these sections. If not specified, Release Engineering & Supplier shall define those in the DVP&R.

NOTE 3: Engineer shall specify objective or subjective acceptance criteria (Example: P99C90, R95C90, No failure or functional degradation, no physical degradation, etc.) in the DVP&R document (reference **QR.00001**.

CONTINUING CONFORMANCE: The Continuing Conformance or CC table below shall include only those tests which are required to ensure on going quality of supplied components. Purpose of CC tests is to identify manufacturing spills before parts leaving the supplier manufacturing locations. Changes resulting from the "Forever Requirements – design or process changes" should not be included.

Table A-2 - Continuing Conformance				
Function (1)	Section #	Sample Size	Acceptance Criteria	Frequency
Appearance	6.1	1	No failures	100%
Appearance	6.1	1	No failures	5/Shift, Note 3
Leak Test	7.1	1	No failures	100%

on December 01, 2025 02:23 AM (UTC+1) - userid: 1728460
DPS-FLA_CHINA@stellantis.com

Table A-2 - Continuing Conformance				
Function (1)	**Section #**	**Sample Size**	**Acceptance Criteria**	**Frequency**
Pull-Apart Effort	7.3	1	No failures	5/Shift
Assembly Effort	7.2	1	No failures	5/Shift
Elevate Temperature Burst Test	6.3.6	10	No failures	1/Month
Dimensional Requirements	6.2.2	1	No failures	Note 2

NOTE 1: Above tests are in addition to all the in-process functional verifications identified in the Control Plans to prevent potential Design & Process failures.

NOTE 2: In accordance with SQA manual Guide to Quality Management Systems Requirement, these characteristics shall be shown to be in statistical control (appropriate process characteristics shall be chosen and monitored using SPC). For circumstances where statistical control cannot be achieved, these characteristics are to be 100% in process tested.

NOTE 3: In addition to the standard appearance checks, supplier must perform CT scan and/or section of molded component to ensure no excessive voids are present. Part should be compared to master sample from initial PPAP.

End of Annex A

on December 01, 2023 02:23 AM (UTC+1) userid: T7284G0
PRS_STLA_CHINA@stellantis.com